

Longer Thomason chains via 3-pole composition

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Abstract

Thomason’s lollipop algorithm turns Smith’s theorem into a procedure: given a Hamiltonian cycle of a cubic graph and one of its edges, it walks through a sequence of Hamiltonian paths and stops at a second Hamiltonian cycle through that edge. The walk can be exponentially long. The best known lower bound, due to Briński and Szady, is $\Omega(1.1812^n)$ steps on 3-connected planar cubic graphs. We construct 3-connected planar cubic graphs with exactly three Hamiltonian cycles on which the algorithm takes $\Theta(\rho^{n/6})$ steps, where $\rho = 2.94771\dots$ is the largest root of $\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1$. Since $\rho^{1/6} = 1.197422\dots$, this is $\Omega(1.19742^n)$.

The construction comes from a general method for analysing the lollipop walk across 3-edge cuts. We prove that, across a 3-edge cut, the walk either behaves exactly as in the graph with one side contracted, or turns back (the *Mirror Lemma*). The side of such a cut away from the start acts as a finite automaton that does not depend on the rest of the graph (the *Transducer Theorem*). For sides in which every pair of ports is joined by a unique Hamiltonian path, the automata never turn the walk back, and nested copies compose by an exact linear recursion. The growth rate of a nested family is therefore the spectral radius of a small integer matrix computed from a constant-size gadget, here a 7-vertex 3-pole. An exhaustive search shows that no nested family of this kind (simple poles, the same gadget at every level) built from a base graph with at most 14 vertices does better.

1 Introduction

Background. Smith’s theorem, published by Tutte [1], states that every edge of a cubic graph lies in an even number of Hamiltonian cycles. Hence a cubic graph with one Hamiltonian cycle has at least three. Thomason [2] gave a constructive proof, the *lollipop algorithm*. Papadimitriou used it as the motivating example for the complexity class PPA [3]: finding a second Hamiltonian cycle in a cubic graph, given one, is a total search problem in PPA. It is not known to be solvable in polynomial time or to be PPA-complete. Eppstein [11] showed that computing the specific cycle returned by the lollipop algorithm lies in $\text{FP}^{\text{PSPACE}}$, and left open whether it is complete for that class.

Exponential lower bounds. Since the lollipop algorithm follows a path in an exponentially large implicit graph, it can be slow. Krawczyk [4] gave the first family of planar cubic graphs on which it takes exponentially many steps; Cameron [5] corrected and generalised the construction. Zhong [6] gave cyclically 4-edge-connected cubic bipartite families with $\Theta(2^{n/16})$ steps, answering a question recorded as Problem 5.5 in [7]. Briński and Szady [8] gave 3-connected planar cubic graphs with $\Omega(1.1812^n)$ steps. This is the best known base [9], and they asked whether families

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with faster growth exist. On the positive side, the walk takes at most n steps on cubic bipartite Pfaffian graphs [9]. A second Hamiltonian cycle in a cubic graph can be found deterministically in polynomial space and $O(1.23103^n)$ time [10].

Our results.

Theorem 1 (main theorem). *Let $\rho = 2.9477115868\dots$ be the largest real root of $\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1$. For every $k \geq 1$ there is a 3-connected planar cubic graph G_k on $n = 8 + 6k$ vertices with exactly three Hamiltonian cycles, together with a Hamiltonian cycle C_k and an edge $v_0v_1 \in C_k$, such that Thomason’s algorithm started from (C_k, v_0v_1) performs exactly $4 + c_{k-1}(\text{ca/B3})$ steps, where $c_{k-1}(\text{ca/B3}) = \Theta(\rho^k)$ is given by an explicit linear recursion (Section 7). Consequently it performs $\Theta(\rho^{n/6})$ steps. Since $\rho^{1/6} = 1.1974227\dots$, this is $\Omega(1.19742^n)$.*

n	14	20	26	32	38	44	50	56	62	68	74	80
steps	10	25	73	214	628	1,849	5,449	16,060	47,338	139,537	411,313	1,212,430

Table 1: Exact numbers of steps of Thomason’s algorithm on G_k from (C_k, v_0v_1) , for $k = 1, \dots, 12$. Consecutive ratios approach $\rho \approx 2.9477$.

Theorem 1 answers the question of Briński and Szady positively for their class of graphs. More important than the constant, perhaps, is the method behind it. It consists of three structural results about the lollipop walk across a 3-edge cut. They hold in every cubic graph, for every 3-edge cut whose edges form a matching and whose side X is away from the start vertex:

- **Mirror Lemma** (Theorem 2). Let X be one side of a 3-edge cut. Then the cycle returned on G projects either to the cycle returned on G/X , or to the input cycle; in the second case the two differ only inside X .
- **Transducer Theorem** (Theorem 4). Every *visit* of the walk to X has an outcome (the walk passes through or turns back), a successor state and a cost, all determined by X and the state at entry alone.
- **Simple poles** (Lemmas 5 and 6). If every pair of ports of X is joined by exactly one Hamiltonian path of X , the walk never turns back at X , and the costs of nested 3-poles satisfy an exact linear recursion.

With these, the analysis of a nested family reduces to a finite computation on its base gadget. The certificate is small enough to be checked by hand (Appendix A). The same method makes it easy to search the design space; within it, our family is optimal for base graphs with at most 14 vertices (Section 8).

Organisation. Section 2 fixes definitions. Section 3 proves the Mirror Lemma, and Section 4 the Transducer Theorem. Section 5 treats simple poles and composition. Section 6 describes the construction, and Section 7 proves Theorem 1. Section 8 reports the search, Section 9 the computational verification, and Section 10 open problems. Appendix A lists every move of the base gadget, and Appendix B traces a complete example.

2 Preliminaries

All graphs are finite and simple. For a vertex v , $N(v)$ is its neighbourhood and $N[v] = N(v) \cup \{v\}$. For $X \subseteq V(G)$, $N(X)$ is the set of vertices outside X with a neighbour in X , and $\delta(X)$ is the set of edges with exactly one end in X . Paths are written as vertex sequences; $\text{rev}(S)$ is the reverse of a sequence S , and $S \cdot T$ is concatenation. For a path P and vertices u, w on it, $P[u..w]$ is the subpath from u to w .

The lollipop walk. Let G be a cubic graph with a Hamiltonian cycle $C = (v_0, v_1, \dots, v_{n-1})$, and fix the edge v_0v_1 . A *state* is a Hamiltonian path of G that starts with v_0v_1 . Let P be a state with endpoint z , and let zw be an edge not on P with $w \neq v_0$. If s is the successor of w on P , the *rotation* of P along zw is

$$P' = P[v_0..w] \cdot \text{rev}(P[s..z]),$$

obtained by adding zw and deleting ws . Rotations are symmetric: P' has endpoint s , and P is the rotation of P' along sw . The *state graph* $\mathcal{G}(G)$ has the states as vertices and rotations as edges. Every state has degree at most 2, since z has two non-path edges. A state whose endpoint is adjacent to v_0 has degree at most 1, and adding the edge to v_0 closes it to a Hamiltonian cycle through v_0v_1 . Conversely, each such cycle arises from exactly one such state.

Thomason's algorithm starts at $S_0 = C - v_{n-1}v_0$, which is a state of degree 1, and follows its component of $\mathcal{G}(G)$. This component is therefore a path, and the algorithm returns the cycle closed by its other end. Operationally, a *step* at endpoint z uses the unique edge zw such that w is neither the predecessor of z nor the *previous attachment* (the w of the previous step; initially v_0). The algorithm stops when this edge goes to v_0 . We count *steps*, i.e. rotations.

3-poles and substitution. A *3-pole* in G is a set $X \subset V(G)$ with $|\delta(X)| = 3$ whose cut edges $e_i = p_iq_i$ ($p_i \in X$, $q_i \notin X$) form a matching. The p_i are the *ports*. The graph G/X obtained by contracting X to a single vertex x is then simple and cubic. Let $h_{ij}(X)$ be the number of Hamiltonian paths of $G[X]$ between p_i and p_j . We call X *simple* if $h_{ij}(X) = 1$ for all $i \neq j$. For a cubic graph F and a vertex r , the graph $F - r$ is a 3-pole with ports $N_F(r)$ once it is placed into a host. For a cubic graph H with a vertex y and a 3-pole P , $H[y \leftarrow P]$ denotes the graph obtained by deleting y and joining its three neighbours to the three ports of P by a fixed bijection, the *wiring*.

Standing hypothesis. Whenever we consider a 3-pole X of a graph in which the lollipop walk runs, we assume $v_0 \notin X \cup N(X)$. In particular $v_1 \notin X$, and the input cycle crosses $\delta(X)$ exactly twice.

3 The Mirror Lemma

Let X be a 3-pole of G , $Y = V(G) \setminus X$, and $H = G/X$. A state starts in Y and alternates between maximal Y -segments and X -segments. Since a Hamiltonian path crosses $\delta(X)$ at most three times, and an even number of times if it ends in Y , every state has exactly one of the following types. We define a map π from states of G to states of H accordingly (Fig. 1).

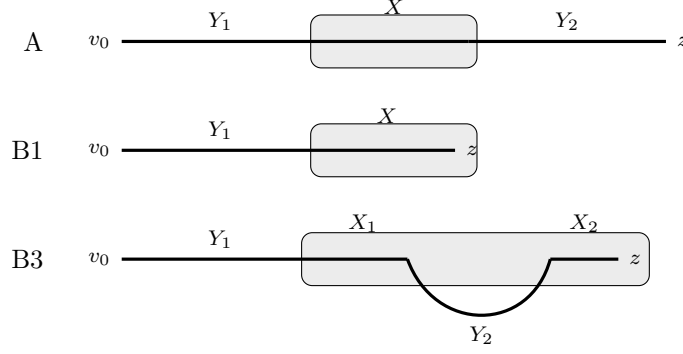


Figure 1: The three types of states relative to a 3-pole X (shaded). Type B3 uses all three cut edges.

type	shape	cut edges used	$\pi(P)$
A	$Y_1 \cdot X \cdot Y_2$, endpoint in Y_2	2	$Y_1 \cdot x \cdot Y_2$
B1	$Y_1 \cdot X$, endpoint in X	1	$Y_1 \cdot x$
B3	$Y_1 \cdot X_1 \cdot Y_2 \cdot X_2$, endpoint in X_2	3	$Y_1 \cdot x \cdot Y_2$

Each $\pi(P)$ is a state of H , since $v_0, v_1 \in Y_1$. For a type B3 state we write p_a for the entry port of X_1 , p_b for its exit port and p_c for the entry port of X_2 . For a type A state, p_{in} and p_{out} are the entry and exit ports of the X -segment.

Theorem 2 (Mirror Lemma). *Under the standing hypothesis, π maps every edge of $\mathcal{G}(G)$ either to an edge of $\mathcal{G}(H)$ or to a single vertex. Consequently, the cycle returned on G projects either to the cycle returned on H , or to C/X . In the second case it agrees with C outside X .*

Proof. Let $P \rightarrow P'$ be the rotation at the endpoint z of P along zw , deleting ws . We use repeatedly that contracting a contiguous block of a sequence commutes with reversing a suffix, provided the cut point ws is not inside the block.

Case A (P of type A, $z \in Y_2$).

- (i) $w, s \in Y$. The X -segment is untouched or reversed as a whole, P' has type A, and $\pi(P')$ is the rotation of $\pi(P)$ along zw .
- (ii) $w \in Y, s \in X$. Then ws is the entry cut edge, so $w = q_{\text{in}}$ is the last vertex of Y_1 and $s = p_{\text{in}}$. Now $P' = Y_1 \cdot \text{rev}(Y_2) \cdot \text{rev}(X)$ has type B1, and $\pi(P') = Y_1 \cdot \text{rev}(Y_2) \cdot x$ is the rotation of $\pi(P)$ along zw , which deletes wx .
- (iii) $w \in X$. Then $zw = e_j$ is the unused cut edge, with $z = q_j$ and $w = p_j$. Since the ports are distinct, $p_j \neq p_{\text{in}}, p_{\text{out}}$ is an interior vertex of the X -segment, so $s \in X$. Writing the X -segment as $X_a \cdot X_b$ with X_a ending at w , we get $P' = Y_1 \cdot X_a \cdot \text{rev}(Y_2) \cdot \text{rev}(X_b)$, of type B3, with $\pi(P') = Y_1 \cdot x \cdot \text{rev}(Y_2)$. In H , the edge $q_j x$ is not on $\pi(P)$, and the rotation of $\pi(P)$ along it deletes the edge from x to the first vertex of Y_2 , giving exactly $\pi(P')$.

Case B1 ($P = Y_1 \cdot X$).

- (i) $w \in X$. Then $s \in X$, the Y -part is unchanged, and $\pi(P') = \pi(P)$.

- (ii) $w \in Y$. Then $z = p_j$ and $w = q_j$ for some $j \neq \text{in}$. As the q_i are distinct, w is not the last vertex of Y_1 , so $s \in Y_1$. Writing $Y_1 = Y_a \cdot Y_b$ with Y_a ending at w , $P' = Y_a \cdot \text{rev}(X) \cdot \text{rev}(Y_b)$ has type A, and $\pi(P') = Y_a \cdot x \cdot \text{rev}(Y_b)$ is the rotation of $\pi(P) = Y_1 \cdot x$ along xq_j . Here $q_j \neq v_0$ by the standing hypothesis.

Case B3 ($P = Y_1 \cdot X_1 \cdot Y_2 \cdot X_2$). All three cut edges are on P , so $w \in X$.

- (i) $w \in X_2$. Then $s \in X_2$ and $\pi(P') = \pi(P)$.
- (ii) $w \in X_1$, $w \neq p_b$. Then $s \in X_1$. Writing $X_1 = X_{1a} \cdot X_{1b}$ with X_{1a} ending at w , $P' = Y_1 \cdot (X_{1a} \cdot \text{rev}(X_2)) \cdot \text{rev}(Y_2) \cdot \text{rev}(X_{1b})$ has type B3. Its first X -piece $X_{1a} \cdot \text{rev}(X_2)$ now ends at the old p_c , so the roles of p_b and p_c are exchanged.
- (iii) $w = p_b$. Then $s = q_b$, and $P' = Y_1 \cdot (X_1 \cdot \text{rev}(X_2)) \cdot \text{rev}(Y_2)$ has type A.

In cases B3(ii) and B3(iii), $\pi(P') = Y_1 \cdot x \cdot \text{rev}(Y_2)$. Since $\pi(P) = Y_1 \cdot x \cdot Y_2$ ends at q_c and $q_c x$ is not on $\pi(P)$, this is the rotation of $\pi(P)$ along $q_c x$, which deletes xq_b .

No other case occurs, because a non-path edge leads from Y into X , or from X into Y , only along an unused cut edge. This proves the first claim.

For the consequence, the walk on G follows the component of $\mathcal{G}(G)$ from $S_0(G)$ to its other end T . Its image under π is a connected subgraph of the component L of $\mathcal{G}(H)$ that contains $\pi(S_0(G)) = S_0(H)$ (here $v_{n-1} \in Y$ by the standing hypothesis), and L is a path whose two ends close to C/X and to the cycle returned on H . The endpoint of T is adjacent to v_0 , so it lies in Y by the standing hypothesis. Hence T has type A, and $\pi(T)$ is a state of L whose endpoint is adjacent to v_0 , i.e. an end of L . Finally, $\pi(T) = S_0(H)$ means that the cycle closed by T uses the same edges as C outside X . $\square$

4 Visits and the Transducer Theorem

Definition 3 (visits). A *visit* to X is a maximal run of consecutive states of the walk whose endpoint lies in X (states of type B1 or B3), together with the step into the run (its *entry*) and the step out of it (its *exit*). By the proof of Theorem 2, a visit starts from a state of type A whose X -segment is a Hamiltonian path Q of X , which we orient from p_{in} to p_{out} . It has one of two *kinds*:

B1: the entry deletes the entry cut edge e_{in} (case A(ii));

B3: the entry adds the unused cut edge at the third port (case A(iii)).

The *cost* of a B1 visit is the number of steps strictly between its entry and its exit. The cost of a B3 visit is the number of steps after its entry, up to and including its exit. The corresponding passage of the walk on H through x takes two steps (B1) or one step (B3), so in both cases the cost is the number of additional steps caused by X . A visit *transmits* if, after it, the projected walk continues as the walk on H does; otherwise it *reflects*.

Theorem 4 (Transducer Theorem). *For every 3-pole X , every oriented Hamiltonian path Q of X between two ports, and every kind $\kappa \in \{\text{B1}, \text{B3}\}$, there are an outcome (transmit or reflect), a successor path Q' and a cost $c_X(Q, \kappa)$, such that every visit of kind κ that starts with X -segment Q , in any host graph satisfying the standing hypothesis, has this outcome, leaves X -segment Q' , and has this cost. More strongly, the whole sequence of steps of the visit, viewed as operations on the X -pieces of the state, depends only on (Q, κ) . Moreover:*

1. *outside visits, every step of the walk on G projects to the step that the rule of Section 2 prescribes on H from the projected state and the corresponding previous attachment;*
2. *after a transmitting visit, the walk on G is at a lift of the state reached by the walk on H , with the same previous attachment, where ports are identified with x ;*
3. *a reflecting visit ends at a lift of the state from which its entry was made.*

Proof. Independence of the host. While the endpoint is in X , a step is determined by the endpoint z , its neighbours, its predecessor and the previous attachment. In type B1 states every attachment lies in X and its successor too, so only the X -segment changes (case B1(i)), until the walk leaves along an unused cut edge (case B1(ii)). In type B3 states the steps are of the forms B3(i), B3(ii), which re-splits the X -pieces and reverses Y_2 as an opaque block, and the exit B3(iii). Hence the sequence of X -pieces, with their order and orientation, evolves as a function of Q and κ alone. After a B3 entry the previous attachment is the port p_j , which lies in X ; in a B1 visit it lies in X after the first internal step. At the first step of a B1 visit it is q_{in} , so that visit cannot leave through p_{in} immediately. The cut edges are part of the data of X , and none of them goes to v_0 , so no other information about the host enters.

Item 1. Outside visits the endpoint z lies in Y . The edges at z , its predecessor and the previous attachment correspond in G and H (a port corresponding to x). So the admissible edge corresponds as well, and by cases A(i)–(iii) the step projects to the step of the walk on H .

Items 2 and 3 for B1. The entry is the step $S \rightarrow S^*$ of the walk on H that makes x the endpoint by deleting $q_{\text{in}}x$, after which the previous attachment is q_{in} . In G the state after the entry is $Y_1 \cdot \text{rev}(Y_2) \cdot \text{rev}(Q)$, which ends at p_{in} and whose only cut edge on the path is e_{out} . (So in the notation of case B1, the entry port of this B1 state is p_{out} , and the exit is through p_{in} or p_c .) The exit rotates along some cut edge p_jq_j and projects to the rotation of S^* along xq_j (case B1(ii)), after which the previous attachment is q_j in both graphs. The walk on H continues from S^* along xq_c , where p_c is the third port. So an exit through p_c is that step (transmit), and an exit through p_{in} is the rotation along xq_{in} , which undoes the entry and returns to S (reflect).

Items 2 and 3 for B3. The entry is the step $S \rightarrow S^*$ of the walk on H that attaches q_j to x , after which the previous attachment is x . Steps of the form B3(ii) project alternately to S and S^* . The exit B3(iii) lands at a lift of $Y_1 \cdot x \cdot \text{rev}(Y_2)$, which is S or S^* , with previous attachment p_b , corresponding to x . At S^* the walk on H continues from there (transmit). At S the endpoint q_j has two non-path edges, q_jx (now forbidden) and one other, so the projected walk goes back along L (reflect).

The cost is the number of steps of the visit, which by the first part depends only on Q and κ . □

5 Simple poles

Lemma 5 (simple poles transmit). *If X is simple, then every visit to X transmits.*

Proof. A state of H of type A (endpoint different from x , with x interior) has exactly one lift to G of type A. Its Y -part is determined, and its X -segment is the unique Hamiltonian path of X between the two ports it uses. (If its endpoint is adjacent to x , it may also have lifts of type B3.) Every visit ends in a state of type A, since both kinds of exit, cases B1(ii) and B3(iii), produce type A. By Theorem 4(3), a reflecting visit ends at a lift of the state from which its entry was made. That state has type A, so the visit ends at its unique type-A lift, which is the state the walk

occupied before the visit. The walk on G would then revisit a state, which is impossible, since it follows a path in $\mathcal{G}(G)$. $\square$

Lemma 6 (composition). *Let P be a 3-pole, and let $X \subset P$ be a simple 3-pole with $N(X) \subseteq P$. Let $P' = P/X$, with X contracted to a vertex x . Then:*

- (a) $h_{ij}(P) = h_{ij}(P')$ for every pair of ports;
- (b) for every visit (Q, κ) to P that occurs in some host satisfying the standing hypothesis, P and P' have the same outcome and successor, under the bijection of (a);
- (c) for every such visit,

$$c_P(Q, \kappa) = c_{P'}(Q/X, \kappa) + \sum_f M_{(Q/X, \kappa), f} c_X(f),$$

where $M_{(Q', \kappa), f}$ counts the passages of kind f through x during the visit (Q', κ) to P' . A passage is a step that makes x the endpoint by deleting x 's entry edge (kind B1), or a step that attaches to x (kind B3), together with the oriented pair of ports of X that the path uses at that moment.

Proof. (a) The ports of P lie outside X : a port of P has a neighbour outside P , which would otherwise lie in $N(X) \setminus P$. So a Hamiltonian path of P between two ports crosses $\delta(X)$ an even number of times, at most three, and at least once, hence exactly twice. It therefore consists of a Hamiltonian path of P' together with a Hamiltonian path of X between the corresponding ports. Since $h_{ij}(X) = 1$, the latter exists and is unique, so this is a bijection.

(b), (c) Fix a host $G = H_0[y \leftarrow P]$ satisfying the standing hypothesis. Since $v_0 \notin P \cup N(P)$ and $N(X) \subseteq P$, the standing hypothesis also holds for X , and $G/X = H_0[y \leftarrow P']$. By Lemma 5 and Theorem 4, the walk on G is the walk on G/X in which each passage through x is replaced by a transmitting visit to X . That visit starts in the X -segment forced by the simplicity of X , and contributes $c_X(f)$ extra steps.

Every passage lies inside a visit to P' , possibly sharing its entry or exit. A B3 passage starts and ends with the endpoint in $N(x) \subseteq P'$. Since x is not a port of P' (by the argument in (a)), a B1 passage either lies inside a visit, or its step is the entry of the visit; the latter happens only for an entry of kind B3 that attaches to a port of P' whose successor is x , so that the entry deletes the edge from that port to x . Its final step, the rotation at x , may be the exit of the visit, of the form B3(iii). In every case the extra steps lie strictly inside the window of the corresponding visit to P . Therefore visits to P in G and visits to P' in G/X correspond one-to-one, with the same kind, starting state up to contraction, outcome and successor, and their costs differ by the inserted costs. The passages during a visit to P' are determined by the visit (Theorem 4), so M does not depend on the host. $\square$

Remark 7. Parts (b) and (c) speak only about visits that occur in some host. If a visit to P occurs in a host G , every passage during it is a visit to X occurring in the same host. In Section 7 we use only visit types that occur, so this is no restriction.

6 The construction

The base graph. Let K be the cubic graph on $\{0, \dots, 7\}$ consisting of the cycle $012 \dots 70$ and the chords $\{0, 4\}, \{1, 3\}, \{2, 6\}, \{5, 7\}$ (Fig. 2). It is 3-connected and planar, and has exactly three

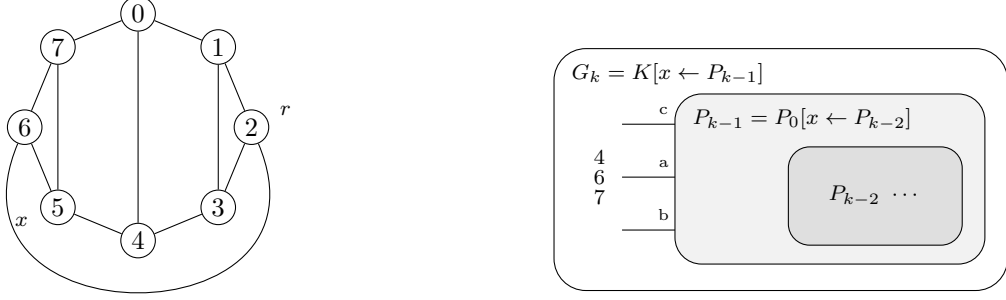


Figure 2: Left: the base graph K , with chords $\{0, 4\}, \{1, 3\}, \{5, 7\}$ inside and $\{2, 6\}$ outside; $x = 5$ is substituted and $r = 2$ deleted. Right: the nesting $G_k = K[x \leftarrow P_{k-1}]$, where the neighbours 4, 6, 7 of x are joined to the ports c, a, b .

Hamiltonian cycles. By exhaustive search, it is the only one of the five connected cubic graphs on 8 vertices with exactly three Hamiltonian cycles. Let $x = 5$ and $r = 2$; then $r \notin N_K[x] = \{4, 5, 6, 7\}$. We call the neighbours of r the ports $a = 1$, $b = 3$, $c = 6$, and fix the *wiring* that joins the neighbours 4, 6, 7 of x to the ports c, a, b of the inner pole, respectively.

The family. Let $G_0 = K$ and $P_0 = K - r$. For $k \geq 1$ let $G_k = K[x \leftarrow P_{k-1}]$ with the wiring above, and $P_k = G_k - r$, where r always denotes the copy of 2 in the outermost copy of K . Then $|V(G_k)| = 8 + 6k$. Since $r \notin N_K[x]$,

$$P_k = (K - r)[x \leftarrow P_{k-1}] = P_0[x \leftarrow P_{k-1}] \quad \text{and} \quad P_k/P_{k-1} = P_0.$$

Let C_k be the Hamiltonian cycle of G_k obtained from $01 \cdots 70$ by replacing x with the unique Hamiltonian path of P_{k-1} between the ports wired to 4 and 6. The walk starts at $v_0 = 0$ with the fixed edge $\{0, 1\}$ of the outermost copy.

Lemma 8. *For every k , the graph G_k is a 3-connected planar cubic graph with exactly three Hamiltonian cycles, and P_k is simple.*

Proof. Simplicity and three cycles. $P_0 = K - r$ is simple. Indeed, each of the three Hamiltonian cycles of K uses one of the three pairs of edges at r ; if pair $\{i, j\}$ is used m_{ij} times, then edge i lies in $m_{ij} + m_{ik}$ cycles, which is even by Smith's theorem and at most 3, and the m 's sum to 3. This forces $m_{ij} + m_{ik} = 2$ for every i , hence $m_{ij} = 1$ for every pair, and $h_{ij}(P_0) = m_{ij} = 1$ (also checked in Table 2). If P_{k-1} is simple, Lemma 6(a), applied to $P_{k-1} \subset P_k$ with $P_k/P_{k-1} = P_0$, shows that P_k is simple. A Hamiltonian cycle of $G_k = K[x \leftarrow P_{k-1}]$ crosses $\delta(P_{k-1})$ exactly twice, so it consists of a Hamiltonian cycle of K , which uses a pair of edges at x , together with a Hamiltonian path of P_{k-1} between the corresponding ports, and this path is unique. Hence G_k has exactly as many Hamiltonian cycles as K , namely three.

3-connectivity. For cubic graphs, 3-connectivity is equivalent to 3-edge-connectivity. Let G and K be 3-edge-connected cubic graphs, $P = G - r$, and suppose F is an edge cut of $K' = K[x \leftarrow P]$ with $|F| \leq 2$ and sides A, B . If $V(P)$ lies on one side, contracting it gives an edge cut of size $|F|$ in K , a contradiction. Otherwise let $F_P = F \cap E(P)$. Placing r on either side of the induced partition of $V(G)$ gives edge cuts of G of sizes $|F_P|$ plus the number of ports on the other side, and both are at least 3. Adding the two inequalities gives $2|F_P| + 3 \geq 6$, so $|F_P| = 2 = |F|$. Then $F \subseteq E(P)$: no wiring edge and no edge of $K - x$ is in F , so each port lies on the same side as its neighbour in K .

visit	cost	passages through x
ab/B1	3	ca/B3
ab/B3	10	ab/B1, ac/B1, ac/B3, bc/B1, bc/B3
ac/B1	5	ab/B3, ac/B1
ac/B3	10	ab/B3, ac/B1, ba/B3, bc/B1, cb/B1
ba/B1	3	ca/B1
ba/B3	10	ab/B3, ac/B1, ca/B3, cb/B1
bc/B1	5	ac/B3, bc/B1
bc/B3	10	ab/B1, ac/B3, bc/B1, cb/B3
ca/B1	3	ba/B1
ca/B3	6	ab/B3, ac/B1
cb/B1	3	cb/B3
cb/B3	6	ac/B3, bc/B1

Table 2: The twelve visit types of $P_0 = K - r$. The first two letters give the oriented pair of ports of the visiting path. All visits transmit, and P_0 is simple. Together the rows define the cost vector c_0 and the passage matrix M .

With $|F_P| = 2$, each of the two inequalities says that the other side contains at least one port, so both sides contain ports, and hence vertices of $K - x$. Placing x on either side therefore gives edge cuts of K with both sides nonempty, whose sizes are the numbers of ports on the other side. Both would have to be at least 3, which is impossible with three ports.

Planarity. Deleting r from a plane embedding of G_{k-1} leaves a face whose boundary meets the three ports in the cyclic order of the rotation at r . Deleting x from a plane embedding of K leaves a face meeting $N_K(x)$ in the cyclic order of the rotation at x . Any bijection between two 3-element sets either preserves or reverses cyclic order, and a reversal is undone by mirroring the embedding of P_{k-1} . So P_{k-1} can be placed inside that face and joined to $N_K(x)$ without crossings. $\square$

7 Proof of the main theorem

The base gadget. The pole $P_0 = K - r$ has six oriented pairs of ports, and hence twelve visit types. Table 2 lists, for each, its cost and its passages through x . All twelve transmit. The passages are named by the ports of the inner pole, through the wiring. Appendix A gives every step.

Let R be the set of visit types reachable from ca/B3 in the directed graph of M (defined below), with an arc $i \rightarrow j$ when $M_{ij} > 0$. It consists of the ten types other than ba/B1 and ca/B1, and it is strongly connected.

Lemma 9. *Every type $i \in R$ occurs as a visit to P_k , for every $k \geq 0$, in the walk on some G_m . For all $k \geq 1$ and $i \in R$, the visit of type i to P_k transmits with the same successor as in P_0 , and*

$$c_k(i) = c_0(i) + \sum_j M_{ij} c_{k-1}(j),$$

where only $j \in R$ contribute.

Proof. By Theorem 4 applied to $X = P_{m-1}$ in G_m , the walk on G_m contains a visit of type ca/B3 to P_{m-1} (see the proof of Theorem 1). If a visit of type i to P_k occurs in the walk on G_m and $M_{ij} > 0$, then the visit contains a passage of type j through the copy of x in P_k , which is a visit of type j to P_{k-1} in the same walk (Lemma 6). By induction on the distance d from ca/B3, a type at distance d occurs as a visit to P_{m-1-d} in G_m for every $m \geq d + 1$; taking $m = k + 1 + d$ gives the

first claim. For the second, apply Lemma 6 to $X = P_{k-1} \subset P = P_k$. This is allowed since P_{k-1} is simple (Lemma 8), $N(P_{k-1}) = N_K(x) \subseteq P_k$, and $P_k/P_{k-1} = P_0$. Since R is closed under the arcs of M , only $j \in R$ contribute. $\square$

With the visit types ordered ab/B1, ab/B3, ac/B1, ac/B3, ba/B1, ba/B3, bc/B1, bc/B3, ca/B1, ca/B3, cb/B1, cb/B3, we have $c_0 = (3, 10, 5, 10, 3, 10, 5, 10, 3, 6, 3, 6)$ and

$$M = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

A direct computation gives the characteristic polynomial

$$\det(\lambda I - M) = \lambda^2(\lambda - 1)^2(\lambda + 1)^2(\lambda^2 + 1)(\lambda^4 - 2\lambda^3 - 2\lambda^2 - 2\lambda - 1).$$

The quartic factor is irreducible over $\mathbb{Q}$, and its largest real root $\rho = 2.9477115868\dots$ is the spectral radius of M , and also of its restriction M_R to R . All other eigenvalues of M have modulus at most 1 (the other roots of the quartic have moduli 0.584 and 0.762).

Proof of Theorem 1. By Lemma 8, G_k is a 3-connected planar cubic graph on $8 + 6k$ vertices with exactly three Hamiltonian cycles.

Exact length. Apply Theorem 4 to the 3-pole $X = P_{k-1}$ of G_k , so $H = G_k/X = K$. The standing hypothesis holds, since $0 \notin N_K[x]$. On K , the walk from C with fixed edge $\{0, 1\}$ takes 4 steps and passes through x once, in a passage of type ca/B3. By Lemma 5 the corresponding visit transmits, so

$$\text{steps}(G_k) = 4 + c_{k-1}(\text{ca/B3}).$$

Lower bound. By Lemma 9, the vector c_k restricted to R satisfies $c_k = c_0 + M_R c_{k-1}$ on R , where M_R is M restricted to R . The integer vector u on R given in the first column of Table 3 satisfies $M_R u \geq \frac{2947}{1000} u$ componentwise; this is a finite check. Let $\varepsilon = \min_{i \in R} c_0(i)/u_i > 0$. Since $c_k \geq M_R c_{k-1}$ on R and $M_R \geq 0$, induction gives $c_k \geq \varepsilon(2947/1000)^k u$ on R . Hence $\text{steps}(G_k) = \Omega(2.947^k)$. The second column of Table 3 is a vector u' with $M_R u' \geq \frac{29477}{10000} u'$, which gives $\Omega(2.9477^k)$ in the same way.

Sharp lower bound. Since R is strongly connected, M_R is irreducible, and by the Perron–Frobenius theorem it has an eigenvector $v > 0$ with $M_R v = \rho(M_R) v = \rho v$. Repeating the argument with v in place of u gives $c_k \geq \varepsilon' \rho^k v$ on R , so $\text{steps}(G_k) = \Omega(\rho^k)$.

Upper bound. On R , $c_k = \sum_{j=0}^k M_R^j c_0$. Since ρ is a simple eigenvalue of M_R and all others have modulus less than ρ , $\|M_R^j\| = O(\rho^j)$, so $c_k = O(\rho^k)$.

Hence $\text{steps}(G_k) = \Theta(\rho^k) = \Theta(\rho^{n/6})$, and $\rho^{1/6} = 1.1974227\dots$ $\square$

type	u	u'
ab/B1	77	746
ab/B3	442	4283
ac/B1	227	2199
ac/B3	442	4283
ba/B3	330	3198
bc/B1	227	2199
bc/B3	330	3198
ca/B3	227	2199
cb/B1	77	746
cb/B3	227	2199

Table 3: Integer vectors with $M_R u \geq 2.947 u$ and $M_R u' \geq 2.9477 u'$ componentwise on the reachable set R .

8 Searching the design space

The method gives the exact growth rate of any nested family $G_k = K[x \leftarrow G_{k-1} - r]$ from the base gadget $K - r$ alone, provided $K - r$ is simple and $r \notin N_K[x]$. Simplicity forces K to have exactly three Hamiltonian cycles. The growth rate per vertex is $\rho(M_R)^{1/(|K|-2)}$, where R is the set of visit types reachable from the passages of the walk on K through x .

We enumerated all Hamiltonian cubic graphs with at most 14 vertices, as a Hamiltonian cycle plus a perfect matching of chords, up to isomorphism. There are 5, 17, 80 and 474 of them on 8, 10, 12 and 14 vertices, in agreement with the known counts. We kept those with exactly three Hamiltonian cycles: there are 1, 3, 7 and 24 of them. For each, we considered all choices of x , $r \notin N[x]$, wiring, Hamiltonian cycle and start. The best per-vertex rates are 1.197423 (8 vertices, our family), 1.184113 (10), 1.191341 (12) and 1.197423 (14). The best 14-vertex family is two levels of ours. So among nested families of simple poles with the same gadget at every level and $|K| \leq 14$, ours is optimal. Mixing gadgets from level to level, so that $c_k = c_0^{(k)} + M^{(k)} c_{k-1}$, gives the growth rate of a periodic word as $\rho(M^{(1)} \dots M^{(p)})^{1/\sum(|K_i|-2)}$. A beam search over periodic words of length up to 8 over all 1,392 distinct gadget matrices with $|K| \leq 12$ found nothing better than 1.197423. Families whose poles are not simple, so that they may reflect the walk, are outside the exact theory. Direct simulation of several thousand of them with $|K| \in \{10, 12\}$ gave at most about 1.189 per vertex.

9 Verification

The computational ingredients of the proof are re-derived by a short standalone script, `verify.py`, supplied with this paper. It uses the Python standard library only and runs in about two seconds. It checks:

- K has exactly three Hamiltonian cycles, and P_0 is simple;
- all twelve visits of P_0 transmit, with the costs and passages of Table 2;
- the reachable set R and both certificates of Table 3, in exact rational arithmetic;
- end to end, for $k \leq 12$: it builds G_k explicitly, runs Thomason’s algorithm from $(C_k, \{0, 1\})$, and compares the number of steps with $4 + c_{k-1}(\text{ca/B3})$, which reproduces every entry of Table 1;

- for $k \leq 3$: cubic, exactly three Hamiltonian cycles (by exhaustive search), and 3-edge-connected.

The remaining computational claims are not needed for Theorem 1 and are checked by separate scripts: the search of Section 8, with exact isomorphism tests (which also shows that K is the only cubic graph on 8 vertices with exactly three Hamiltonian cycles), and the direct checks of the graph class for larger n reported below. We also tested the ingredients independently of this script. An edge-set implementation of the algorithm and a brute-force traversal of the full state graph agree with it on all tested graphs. A second implementation written from the definitions in this paper alone reproduces Table 1 (and gives 3,573,892 steps for $k = 13$, as the recursion predicts), Appendix B, and Table 2 together with Appendix A. The latter it derives from 109,632 runs of the walk on 21 different hosts $H[y \leftarrow P_0]$, in which every visit type behaves identically. The Mirror Lemma and the Transducer Theorem were checked on thousands of random hosts and 3-poles. Lemma 6 was checked on 71,484 random pairs (P', X) , and Lemma 5 on 7,233 random simple poles with up to 21 vertices, without exception. When X is not simple, the conclusions of Lemma 6 fail in most cases, so the hypothesis cannot be dropped. Planarity and 3-connectivity were confirmed directly for $n \leq 80$, and the number of Hamiltonian cycles by exhaustive search for $n \leq 56$.

10 Discussion and open problems

The Mirror Lemma and the Transducer Theorem show that a 3-edge cut behaves like a reversible finite-state gadget that passes the walk through or reflects it. The lollipop walk is then a deterministic reversible particle moving through such gadgets, which is the setting of zero-player reversible motion planning, where PSPACE-hardness results are known [12]. Simple poles never reflect, so they cannot carry information by themselves. Poles with several internal paths can reflect, and their state persists between visits. We believe this is a possible route to Eppstein’s question [11] whether computing the output of the lollipop algorithm is $\text{FP}^{\text{PSPACE}}$ -complete.

- Is there an exact composition theory for non-simple poles, and do such poles give larger bases?
- Does the method extend to 4-edge cuts? This would be needed for cyclically 4-edge-connected graphs, the class of Zhong’s examples and of Thomassen’s reduction.
- Is there an upper bound of the form c^n with $c < 2$ on the length of the lollipop walk in cubic graphs?
- Is the lollipop walk polynomial on cyclically 5-edge-connected cubic graphs?

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A Every step of the twelve visits of P_0

Vertices carry their labels in K . The column Q lists the internal path of the visit, from the port nearer v_0 . A B1 visit starts with the endpoint at the first vertex of Q , with Q reversed. A B3 visit starts after the cut edge at the third port has been added; the internal path is then split after that port. Each step zw rotates at the endpoint z along the edge zw . † marks a step that makes $x = 5$ the endpoint (a B1 passage), and ‡ a step that attaches to x (a B3 passage). A † on the visit name means that its entry is itself a B1 passage. Every visit transmits.

visit	Q	steps after the entry	cost	passages through x
ab/B1	1076543	13 40 75 †	3	ca/B3
ab/B3 †	1076543	57 65 ‡ 40 76 † 54 31 07 † 56 75 ‡ 43	10	ab/B1, ac/B1, ac/B3, bc/B1, bc/B3
ac/B1	1340756	10 45 ‡ 76 † 57 04	5	ab/B3, ac/B1
ac/B3	1340756	45 ‡ 76 † 57 01 34 † 56 70 45 ‡ 67 † 56	10	ab/B3, ac/B1, ba/B3, bc/B1, cb/B1
ba/B1	3456701	31 04 † 57	3	ca/B1
ba/B3	3456701	75 ‡ 67 04 † 56 70 13 45 ‡ 76 † 57 01	10	ab/B3, ac/B1, ca/B3, cb/B1
bc/B1	3104576	34 07 † 56 75 ‡ 40	5	ac/B3, bc/B1
bc/B3	3104576	07 † 56 75 ‡ 43 10 76 † 54 07 65 ‡ 76	10	ab/B1, ac/B3, bc/B1, cb/B3

visit	Q	steps after the entry	cost	passages through x
ca/B1	6570431	67 [†] 54 01	3	ba/B1
ca/B3	6570431	10 45 [†] 76 [†] 57 04 31	6	ab/B3, ac/B1
cb/B1	6754013	65 [†] 70 43	3	cb/B3
cb/B3	6754013	34 07 [†] 56 75 [†] 40 13	6	ac/B3, bc/B1

B A worked example: the walk on G_1

$G_1 = K[x \leftarrow P_0]$ has 14 vertices. Outer vertices are written $0, \dots, 7$ (without 5), and the inner copy of $K-r$ as $0', 1', 3', \dots, 7'$. The wiring joins $4-6'$, $6-1'$ and $7-3'$. The walk starts from C_1 with fixed edge $\{0, 1\}$. Steps marked $\bullet$ have their endpoint inside the inner pole. Step 1 is the entry of a B3 visit of type ca/B3, and steps 2–7 are its cost $c_0(\text{ca/B3}) = 6$, ending at step 7. The total is $4 + 6 = 10$ steps, as predicted.

step	rotation (z, w)	path after the step
0	–	0 1 2 3 4 6' 5' 7' 0' 4' 3' 1' 6 7
1	$(7, 3')$	$\bullet$ 0 1 2 3 4 6' 5' 7' 0' 4' 3' 7 6 1'
2	$(1', 0')$	$\bullet$ 0 1 2 3 4 6' 5' 7' 0' 1' 6 7 3' 4'
3	$(4', 5')$	$\bullet$ 0 1 2 3 4 6' 5' 4' 3' 7 6 1' 0' 7'
4	$(7', 6')$	$\bullet$ 0 1 2 3 4 6' 7' 0' 1' 6 7 3' 4' 5'
5	$(5', 7')$	$\bullet$ 0 1 2 3 4 6' 7' 5' 4' 3' 7 6 1' 0'
6	$(0', 4')$	$\bullet$ 0 1 2 3 4 6' 7' 5' 4' 0' 1' 6 7 3'
7	$(3', 1')$	0 1 2 3 4 6' 7' 5' 4' 0' 1' 3' 7 6
8	$(6, 2)$	0 1 2 6 7 3' 1' 0' 4' 5' 7' 6' 4 3
9	$(3, 1)$	0 1 3 4 6' 7' 5' 4' 0' 1' 3' 7 6 2
10	$(2, 3)$	0 1 3 2 6 7 3' 1' 0' 4' 5' 7' 6' 4